

Excel Data Analysis

Veda Technologies – Data Analytics Internship Task 5 – Retail Sales Data Analysis using Microsoft Excel

1. Executive Summary

This report documents Task 5 of the Veda Technologies Data Analytics Internship. The task focused on analyzing a structured Retail Sales dataset using Microsoft Excel. Five PivotTables were created to examine regional sales, category-level sales, category-level profit, regional profit, and the relationship between total sales and profit.

2. Task Objective

The objective was to become comfortable analyzing a structured dataset using Excel's core analytical features. The work focused on using PivotTables, sorting, filtering and aggregation to answer practical business questions.

3. Tools and Dataset

Tool: Microsoft Excel

Dataset: Retail Sales Dataset

Techniques: PivotTables, sorting, filtering, aggregation and comparative analysis.

4. PivotTable 1 – Sum of Sales by Region

Business Question: Which regions contribute to overall sales performance?

This PivotTable groups sales by region and calculates total sales for each region, making regional sales contributions easier to compare.

Takeaway: Regional sales comparison helps identify differences in sales contribution across the regions represented in the dataset.

5. PivotTable 2 – Sum of Sales by Category

Business Question: Which product categories generate higher sales?

This PivotTable summarizes total sales for each product category and provides a clear category-level comparison.

Takeaway: Category-level sales analysis makes it easier to identify categories with stronger or weaker sales contributions.

6. PivotTable 3 – Profit by Category

Business Question: Which product categories contribute more to profitability?

This analysis focuses on profit rather than sales volume. It provides an additional perspective on category-level business performance.

Takeaway: Comparing profit by category provides a different perspective from sales and helps evaluate category-level profitability.

7. PivotTable 4 – Profit by Region

Business Question: Which regions generate higher profits?

This PivotTable summarizes profit across regions and can be compared with regional sales to understand differences between sales and profitability.

Takeaway: Regional profit analysis helps distinguish sales performance from profitability.

8. PivotTable 5 – Sum of Sales and Profit

Business Question: How do overall sales and profit compare?

This PivotTable brings sales and profit together in one summary, providing a broader view of retail performance.

Takeaway: Viewing sales and profit together provides a more complete picture of overall business performance.

9. Analysis Workflow

Retail Sales Dataset → Data Preparation → PivotTables → Sales Analysis → Profit Analysis → Business Interpretation.

The analysis keeps the original data separate from the working and summary sections so the source remains available for reference.

10. Key Observations

- Regional sales can be compared to understand differences in sales contribution.
- Category-level analysis helps identify stronger and weaker sales contributors.
- Profit analysis adds an important perspective beyond sales volume.
- Regional profitability can be compared independently from regional sales.
- Viewing sales and profit together provides a broader understanding of business performance.

11. Skills Practiced

This task provided hands-on practice with Microsoft Excel, PivotTables, aggregation, sorting and filtering, sales analysis, profit analysis, and translating numerical summaries into business-oriented observations.

12. Interview Questions

SUMIFS vs PivotTable: SUMIFS is useful for a specific worksheet calculation, while PivotTables are useful for flexible multi-dimensional summaries.

Keeping PivotTables live: Converting source data into an Excel Table allows new rows to be included in the source range, followed by refreshing the PivotTable.

Retail KPI: Total Sales is useful for monitoring sales volume, while Profit or Profit Margin provides a profitability perspective.

13. Deliverables

- Excel workbook containing the Retail Sales analysis.
- Five PivotTables answering different business questions.
- Sales and profit comparisons across regions and categories.
- Summary of analytical observations.

14. Conclusion

This task demonstrates the practical use of Microsoft Excel for structured data analysis. By creating five PivotTables, the Retail Sales dataset was examined from regional, category, sales, and profit perspectives. The task strengthened the ability to summarize data efficiently and use those summaries to answer meaningful business questions.

Internship: Veda Technologies – Data Analytics Internship

Task: Task 5 – Excel Data Analysis

Tool: Microsoft Excel

Dataset: Retail Sales Dataset